

# FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

## PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) with a B.Tech in IT and an AI/ML engineering internship delivering complex geospatial ML models and evaluation systems at national scale. Builds Generative AI tooling — prompt engineering, retrieval augmented generation and agent-style MCP delegation — alongside tree-based predictive models. Seeking an Associate Data Scientist role in the AI & Data Science Centre of Excellence at Suncorp, Melbourne.

## TECHNICAL SKILLS

**Languages:** Python (primary), SQL, TypeScript, JavaScript, Bash

**Machine learning:** XGBoost, scikit-learn, predictive analytics, evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction), pandas, OpenCV

**Generative AI:** prompt engineering, retrieval augmented generation, agents (MCP servers, model routing), LLM fine-tuning (Unsloth, QLoRA, GGUF), Ollama, Claude Code / Cursor, Anthropic API

**Engineering:** Git / GitHub, VS Code, Colab / Jupyter, Docker (basic), Kestra, Linux, REST/JSON

**Also:** GeoPandas, Shapely, QGIS

## EXPERIENCE

### AI/ML Engineering Intern — Prodegee

Mar 2026 – Jun 2026

Remote, Melbourne • RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Ran footprint inference across 200 villages in 7 Mauritanian Wilayas and trained an XGBoost multi-output tier regressor; documented that its 0.816 "confidence" measured agreement with the training heuristic, not accuracy.
- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Built a  $\pm 20\%$  sensitivity harness that identified cluster radius as the dominant parameter ( $\pm 36\text{--}44\%$  swing across 191 villages), driving a 75  $\rightarrow$  200 m recalibration that fixed a 90%+ household under-count (0.069  $\rightarrow$  0.415).
- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a  $\sim 35\times$  over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13%  $\rightarrow$  37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries; authored cross-team handoff contracts.
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and wrote the semester-2 handoff memo.
- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.

**Tools:** Python, XGBoost, scikit-learn, pandas, GeoPandas, Shapely, Git, VS Code, Kestra, Claude Code (MCP, hooks)

### Dispatch Team Member — Amart Furniture

Feb 2025 – Mar 2026

Melbourne, VIC • part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

## EDUCATION

### Master of Artificial Intelligence — RMIT University

Jul 2024 – Oct 2026

Melbourne, VIC • Coursework: Machine Learning, Computer Vision, Intelligent Decision Making (ASP/Clingo), AI Systems Design

### Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology

Dec 2020 – May 2024

Hyderabad, India (affiliated to JNTUH) • Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

## KEY PROJECTS

### Gemma Delegation System — local/cloud LLM hybrid • Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab

May 2026

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model, failing open to the frontier model whenever Ollama is unavailable so delegation never blocks the user.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset; Colab then fine-tunes confirmed examples (4-bit QLoRA, LoRA  $r=16$ ) and exports GGUF back into Ollama.

↳ GitHub: [github.com/faraaz817/gemma-delegation-system](https://github.com/faraaz817/gemma-delegation-system)

- **Agent skills for Claude Code / Cursor:** authored five reusable skills — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs. [github.com/faraaz817/My\\_Skills](https://github.com/faraaz817/My_Skills)

**Bro — AI portfolio chatbot** • Next.js, TypeScript, Anthropic API

2026

- Built a recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; deployed on Vercel.

↳ Live: [job-go-mauve.vercel.app/portfolio](https://job-go-mauve.vercel.app/portfolio)

**Fire Detection & Alerting System** • Python, OpenCV, Haar cascades, threading

2023 – 24

B.Tech capstone, team of 4 — real-time detection from a live camera feed, with off-site alerting

- Trained a custom OpenCV Haar cascade end to end and chose it over heavier detectors on latency grounds — reaction time beats a few points of accuracy for a spreading fire.
- Built the real-time loop: per-frame detection, bounding-box overlay, and threaded alarm and WhatsApp alerting so detection and notification run without blocking each other.

## PUBLICATION

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K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers (40 combinations) on a 15,000-image, four-class maize dataset; best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn). ↳ [questjournals.org/jses](https://questjournals.org/jses) • Code: [github.com/faraaz817/Machine\\_Learning](https://github.com/faraaz817/Machine_Learning)

## CERTIFICATIONS, AWARDS & LEADERSHIP

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- IBM Z Xplore – Concepts and IBM Z & LinuxONE Community Contributor – Level 1 (IBM digital badges via Credly, 2026).
- Smart India Hackathon delegate, Government of India (2022) • Certificate of Publication, Quest Journals (2023).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- Volunteer fundraiser, Taher Foundation (2023).

## LANGUAGES

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English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

## REFEREES

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Available on request.